WEB FUNDAMENTALS

Report Structure

- 1. Objective
- 2. Concept/Understanding
- 3. Practical Work
- 4. Tools Used
- 5. Evidence
- 6. Result
- 7. Key Learnings

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1. Objective

The objective of this task was to understand how web applications work and how HTTP communication happens between a browser and a web server. This covers web application architecture, HTTP/HTTPS, requests and responses, HTTP methods, headers, cookies and sessions, parameters, authentication, status codes, and basic APIs.

2. Concept/Understanding

Web Application Basics

A web application has three main parts:

- **Client** — The browser, such as Chrome or Firefox, that sends requests
- **Server** — The computer that hosts the website and responds
- **Database** — Where all data is stored

Request-Response Flow

When you visit a website:

1. The browser sends a request to the server
2. The server processes the request
3. The server sends back a response
4. The browser displays the page

HTTP in Detail

HTTP (HyperText Transfer Protocol) is how browsers and servers communicate.

HTTP vs HTTPS

- **HTTP** — Unencrypted
- **HTTPS** — Encrypted with TLS

HTTP Methods

Method	Purpose
GET	Retrieve data
POST	Submit data
PUT	Update data
DELETE	Remove data

Headers

Headers are metadata sent with requests and responses. Examples include:

- **User-Agent** — Tells the server which browser you use
- **Content-Type** — Tells what type of data is sent
- **Cookie** — Sends stored data back to the server

Status Codes

Status Code	Meaning
200	OK
301	Moved Permanently
403	Forbidden
404	Not Found
500	Internal Server Error

Cookies and Sessions

- **Cookies** — Small data stored in the browser
- **Sessions** — State maintained on the server

Parameters

Parameters are data sent in the URL with GET requests or in the body with POST requests.

Authentication

Authentication is the process of verifying who you are.

Basic APIs

Applications communicate using APIs that send and receive data.

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3. Practical Work

I completed two TryHackMe rooms:

- Web Application Basics
- HTTP in Detail

I also used browser Developer Tools (F12) to inspect:

- Network requests
- Request and response headers
- Cookies
- HTTP status codes

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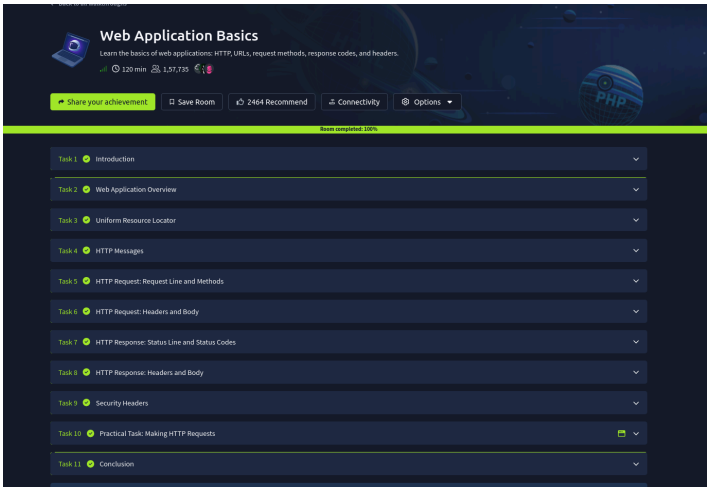
4. Tools Used

- TryHackMe
- Google Chrome
- Browser Developer Tools (F12)

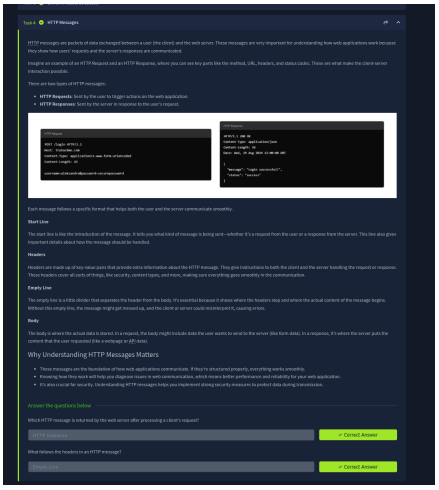
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5. Evidence

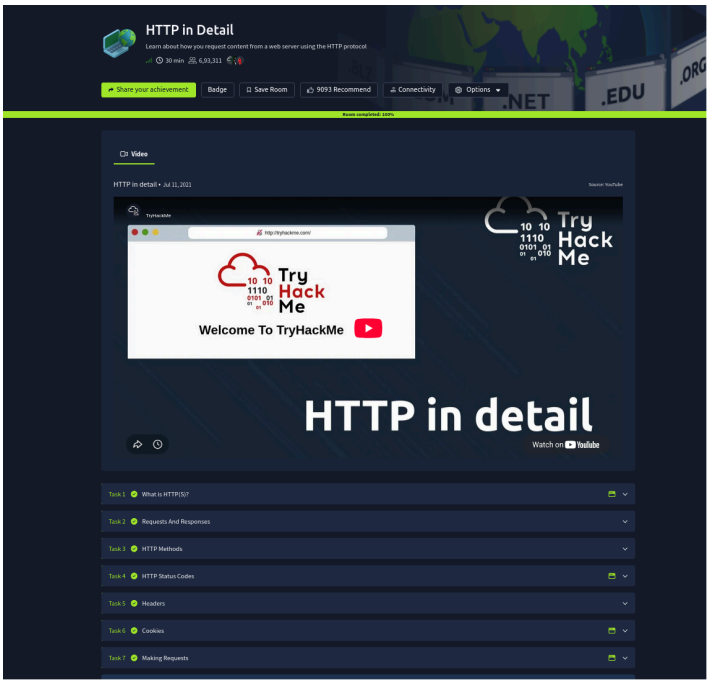
Web Application Basics room completion



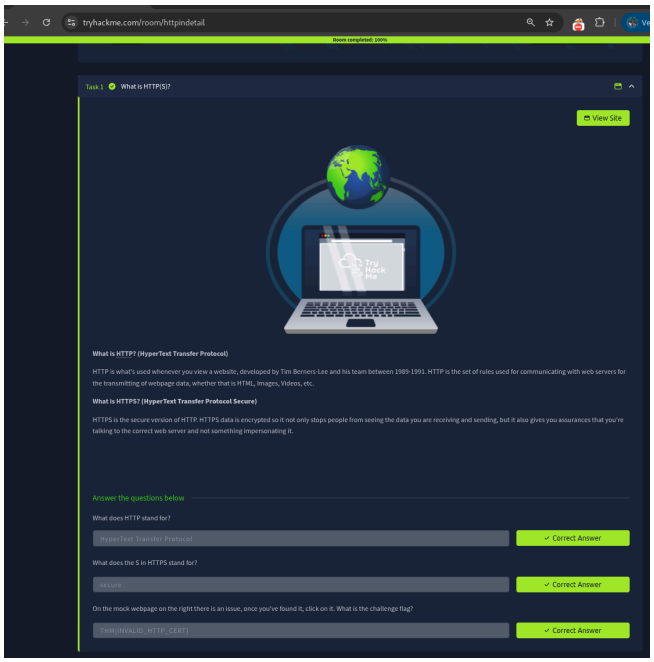
Correct answers in Web Application Basics



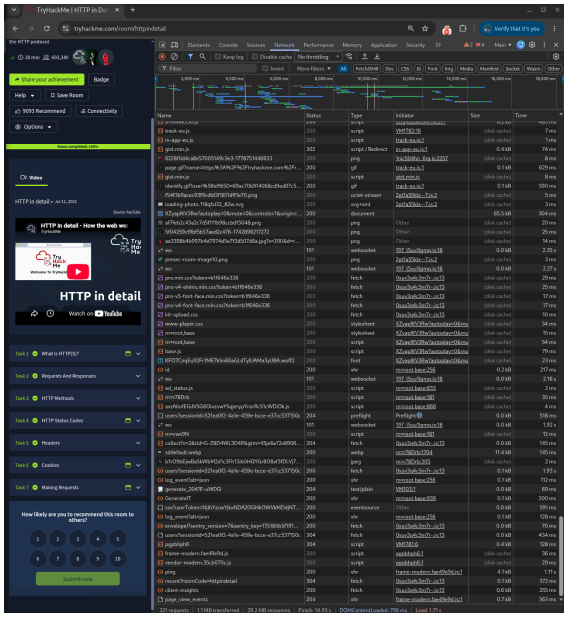
HTTP in Detail room completion



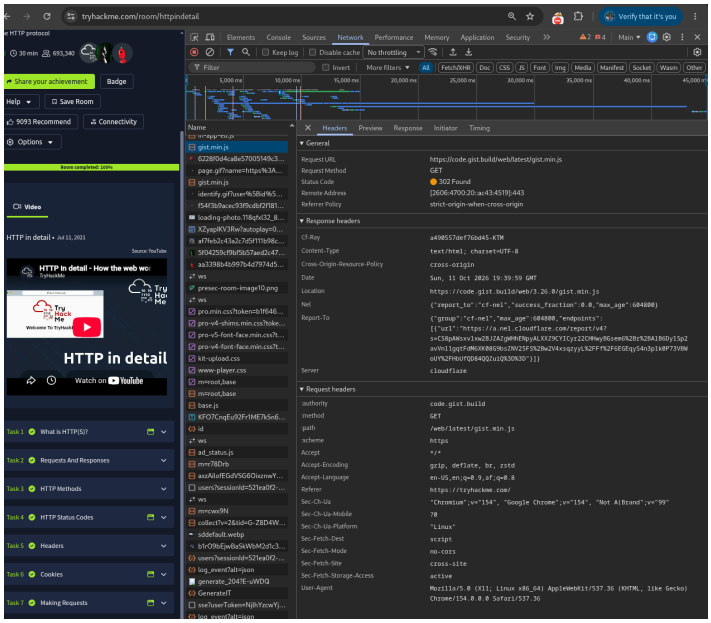
Correct answers in HTTP in Detail



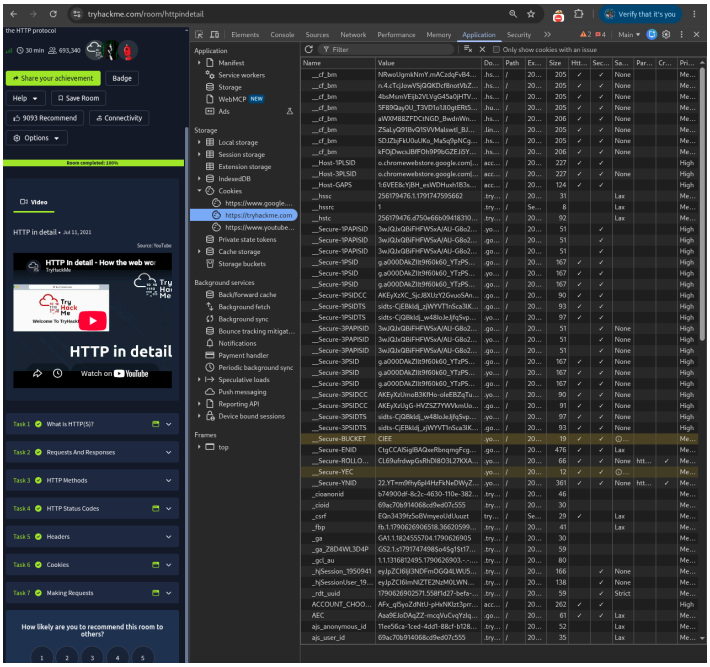
Network tab showing HTTP requests



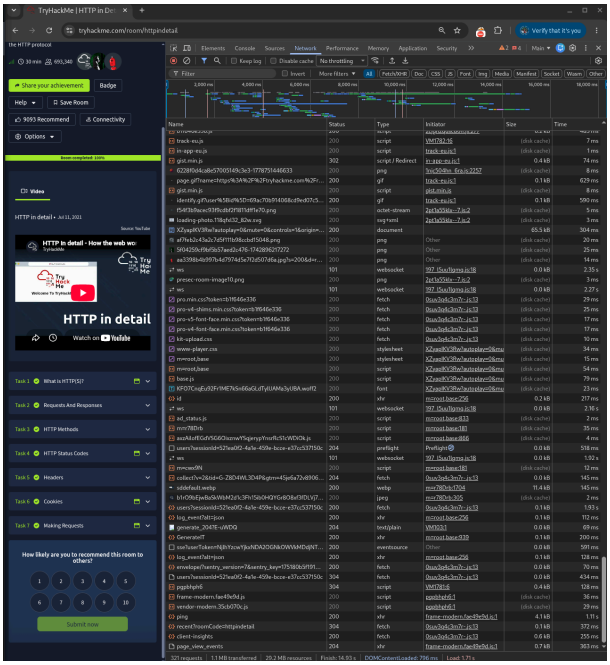
Request and Response headers



Cookies stored in browser



HTTP status codes



6. Result

I successfully understood:

- How web applications are structured
- How HTTP requests and responses work
- Differences between HTTP methods
- The role of headers in communication
- How cookies and sessions work
- How to inspect web traffic using Developer Tools
- HTTP status code meanings

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7. Key Learnings

- Every website uses HTTP to communicate
- HTTP is a stateless protocol
- Cookies and sessions maintain state
- Headers carry important metadata
- Status codes tell the outcome of a request
- Developer Tools reveal everything in HTTP
- Understanding HTTP is the first step in web security
